Detailed Calculation Breakdown

Topological Magnon Band Structure in $Y_2V_2O_7$:
Weyl Magnons and Polarimetric RIXS Signatures

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1 Overview and Scope

This document provides a detailed breakdown of all theoretical calculations presented in the manuscript "Topological Magnon Band Structure in $Y_2V_2O_7$: Weyl Magnons and Polarimetric RIXS Signatures." Each calculation is explained step-by-step, with explicit attention to:

- Physical assumptions and their justifications
- Mathematical derivations and approximations
- Numerical implementation details
- Parameter choices and their sources
- Potential limitations and caveats

The calculations fall into four main categories:

1. **Linear Spin-Wave Theory (LSWT)** — Magnon band structure
2. **Topological Analysis** — Berry curvature and Chern numbers
3. $k \cdot p$ **Theory** — Weyl point characterization
4. **Cluster Exact Diagonalization (CED)** — RIXS spectra

2 Spin Hamiltonian Construction

2.1 The Full Hamiltonian

The starting point is the spin Hamiltonian for nearest-neighbor V^{4+} ions on the pyrochlore lattice:

$$\boxed{\mathcal{H} = -J \sum_{\langle ij \rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + \sum_{\langle ij \rangle} \boldsymbol{D}_{ij} \cdot (\boldsymbol{S}_i \times \boldsymbol{S}_j)} \tag{1}$$

Term-by-term breakdown

Heisenberg Exchange (First Term):

$$\mathcal{H}_{\text{Heis}} = -J \sum_{\langle ij \rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j = -J \sum_{\langle ij \rangle} \left(S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z \right) \tag{2}$$

The sign convention: $J > 0$ means *ferromagnetic* coupling (parallel spins lower energy). This is opposite to the antiferromagnetic convention where $J > 0$ favors antiparallel alignment.

Dzyaloshinskii-Moriya (DM) Interaction (Second Term):

$$\mathcal{H}_{\text{DM}} = \sum_{\langle ij \rangle} \boldsymbol{D}_{ij} \cdot (\boldsymbol{S}_i \times \boldsymbol{S}_j) \tag{3}$$

Expanding the cross product:

$$\boldsymbol{S}_i \times \boldsymbol{S}_j = \begin{pmatrix} S_i^y S_j^z - S_i^z S_j^y \\ S_i^z S_j^x - S_i^x S_j^z \\ S_i^x S_j^y - S_i^y S_j^x \end{pmatrix} \tag{4}$$

The DM interaction is **antisymmetric**: $\boldsymbol{D}_{ij} = -\boldsymbol{D}_{ji}$, so the ordering of sites i, j matters.

2.2 DM Vector Constraints from Symmetry

Moriya's Rules Applied to Pyrochlore

In the pyrochlore structure ($Fd\bar{3}m$, No. 227):

- Each nearest-neighbor bond lies on a C_2 axis
- The bond midpoint is the inversion center for that axis
- $\boldsymbol{D}_{ij}$ must be perpendicular to the bond direction
- $\boldsymbol{D}_{ij}$ lies in the plane containing the bond and the C_2 axis

The six DM vectors on a single tetrahedron are related by C_3 rotation about [111]. Only the magnitude $D = |\boldsymbol{D}_{ij}|$ is a free parameter.

2.3 Parameter Values and Sources

Parameter	Value	Units	Source
J (exchange)	8.22	meV	Onose et al. 2010 (INS on $Lu_2V_2O_7$)
D/J (DM ratio)	0.32	–	Thermal Hall fit ($Lu_2V_2O_7$)
D (DM strength)	2.63	meV	$D = 0.32 \times J$
a (lattice constant)	9.89	Å	Gardner et al. 2010
T_C (Curie temp.)	70	K	Ideue et al. 2012

Table 1: Model parameters and their experimental sources.

Parameter Transfer Assumption

Critical caveat: No direct measurements of J or D exist for $Y_2V_2O_7$ itself. Parameters are *transferred* from $Lu_2V_2O_7$ based on:

- Similar $T_C \approx 70$ K
- Same V^{4+} ($3d^1$, $S = 1/2$) electronic configuration
- Similar lattice constants (9.89 vs 9.93 Å)

Uncertainty: The V–O–V bond angle differs by 1–2° due to different A-site ionic radii (Y^{3+} vs Lu^{3+}), potentially shifting D/J by ±20%.
Robustness: The Weyl crossing persists over $0.1 \lesssim D/J \lesssim 0.6$, so the topology is robust to this uncertainty.

3 Holstein-Primakoff Transformation

3.1 The Transformation

To map the spin problem to a bosonic magnon problem, we use the Holstein-Primakoff (HP) transformation:

Holstein-Primakoff Mapping

For a spin S at site i, define bosonic operators $a_i, a_i^\dagger$ satisfying $[a_i, a_j^\dagger] = \delta_{ij}$. The exact transformation is:

$$S_i^+ = \sqrt{2S}\sqrt{1 - \frac{a_i^\dagger a_i}{2S}}\, a_i \tag{5}$$

$$S_i^- = \sqrt{2S}\, a_i^\dagger \sqrt{1 - \frac{a_i^\dagger a_i}{2S}} \tag{6}$$

$$S_i^z = S - a_i^\dagger a_i \tag{7}$$

where $S^\pm = S^x \pm iS^y$.

Linear Spin-Wave Approximation: Expand the square root to leading order in $1/S$:

$$\sqrt{1 - \frac{n_i}{2S}} \approx 1 - \frac{n_i}{4S} - \frac{n_i^2}{32S^2} - \dots \tag{8}$$

Keeping only the leading term ($n_i/S \ll 1$, i.e., low magnon density):

$$\boxed{S_i^+ \approx \sqrt{2S}\, a_i, \quad S_i^- \approx \sqrt{2S}\, a_i^\dagger, \quad S_i^z = S - a_i^\dagger a_i} \tag{9}$$

This is **exact** for $S \to \infty$ and an approximation for finite S.

Validity for $S = 1/2$

Critical caveat: V^{4+} has $S = 1/2$, the *smallest possible* spin. LSWT corrections scale as $1/S$, so quantum corrections are expected to be $\sim 100\%$ nominally.

Mitigating factors:

- Prior $1/S$ studies of $Lu_2V_2O_7$ show actual renormalization is $\sim 20\%$ due to partial cancellation
- The Weyl crossing is *topologically protected* by C_3 symmetry — interactions can shift t_W and ω_W but *cannot gap* the crossing

3.2 Substitution into the Hamiltonian

LSWT Hamiltonian Construction

Substituting the HP transformation into $\mathcal{H}$:
Heisenberg term:

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+) \tag{10}$$
$$= (S - n_i)(S - n_j) + S(a_i a_j^\dagger + a_i^\dagger a_j) \tag{11}$$
$$\approx S^2 - S(n_i + n_j) + S(a_i a_j^\dagger + a_i^\dagger a_j) + O(a^4) \tag{12}$$

The constant S^2 is the classical ground state energy. The linear terms vanish for the ferromagnetic ground state (no sublattice canting). The bilinear terms give the magnon hopping.
DM term: Similarly generates off-diagonal magnon hopping with complex (imaginary) matrix elements.
The full bilinear Hamiltonian in k-space is:

$$\mathcal{H}_{\text{SW}} = \frac{1}{2} \sum_{\boldsymbol{k}} \boldsymbol{\Psi}_{\boldsymbol{k}}^\dagger \begin{pmatrix} A(\boldsymbol{k}) & B(\boldsymbol{k}) \\ B^*(-\boldsymbol{k}) & A^T(-\boldsymbol{k}) \end{pmatrix} \boldsymbol{\Psi}_{\boldsymbol{k}} \tag{13}$$

where $\boldsymbol{\Psi}_{\boldsymbol{k}}^\dagger = (a_1^\dagger(\boldsymbol{k}), \ldots, a_4^\dagger(\boldsymbol{k}), a_1(-\boldsymbol{k}), \ldots, a_4(-\boldsymbol{k}))$ is the Nambu spinor for 4 sublattices.

Simplification for Collinear Ferromagnet

For the collinear ferromagnetic ground state (all spins parallel along a global axis):

- The anomalous terms vanish: $B(\boldsymbol{k}) = 0$
- The 8×8 BdG problem reduces to a 4×4 Hermitian eigenvalue problem

This simplification occurs because HP expansion in a uniform frame produces only number-conserving hopping.

4 Bogoliubov Diagonalization

4.1 The Colpa Algorithm

Paraunitary Diagonalization

The BdG Hamiltonian must be diagonalized while preserving the bosonic commutation relations. This requires a *paraunitary* transformation $\mathcal{T}(\boldsymbol{k})$ satisfying:

$$\mathcal{T}^\dagger \Sigma_z \mathcal{T} = \Sigma_z \tag{14}$$

where $\Sigma_z = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1)$ is the bosonic metric.
Colpa's Algorithm (1978):

1. Form $\Sigma_z H(\boldsymbol{k})$ where H is the BdG matrix

2. Compute eigenvalues and eigenvectors of $\Sigma_z H$
3. Eigenvalues come in $\pm$ pairs; positive ones are magnon energies
4. Normalize eigenvectors with respect to the Σ_z metric

For our collinear FM case with $B = 0$, this reduces to standard Hermitian diagonalization of the 4×4 matrix $A(\boldsymbol{k})$.

4.2 Numerical Implementation Details

- High-symmetry path: $\Gamma \to X \to W \to L \to \Gamma$ (FCC convention)
- Sampling: 300 points per segment
- Weyl point search: 5000 points along $\Gamma \to L$, gap threshold 0.05 meV
- Magnon Hall conductivity: $20 \times 20 \times 20$ Monkhorst-Pack grid

5 Berry Curvature Calculation

5.1 Definition and Physical Meaning

The Berry curvature is the "magnetic field" in k-space:

$$\boldsymbol{\Omega}_n(\boldsymbol{k}) = i\langle \nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}}| \times |\nabla_{\boldsymbol{k}} u_{n\boldsymbol{k}}\rangle \tag{15}$$

Kubo Formula for Berry Curvature

A gauge-invariant expression uses the velocity matrix elements:

$$\boxed{\Omega_n^{\alpha\beta}(\boldsymbol{k}) = -2\,\mathrm{Im} \sum_{m \neq n} \frac{\langle n\boldsymbol{k}|\partial_{k_\alpha}\mathcal{H}|m\boldsymbol{k}\rangle\langle m\boldsymbol{k}|\partial_{k_\beta}\mathcal{H}|n\boldsymbol{k}\rangle}{[E_m(\boldsymbol{k}) - E_n(\boldsymbol{k})]^2}} \tag{16}$$

Why gauge-invariant? Depends only on:

- Energy differences $E_m - E_n$ (gauge-invariant)
- Off-diagonal velocity matrix elements (invariant under phase changes of eigenstates)

Numerical implementation:

- Derivatives $\partial_{k_\alpha}\mathcal{H}$ computed by finite differences
- Step size: $\delta k = 2 \times 10^{-5}$ Å^{-1}
- Sum over m includes all three other magnon branches

5.2 Berry Curvature Near a Weyl Point

Monopole Structure

A Weyl point acts as a *Berry monopole* in k-space:

$$\boldsymbol{\Omega}_n(\boldsymbol{k}) \propto \frac{\hat{\boldsymbol{r}}}{|\boldsymbol{k} - \boldsymbol{k}_W|^2} \tag{17}$$

This $1/r^2$ divergence is the signature of a topological defect. The color map in Fig. 7a of the paper shows this monopole structure.

6 Chern Number and Topological Charge

6.1 Definition

$$\boxed{\mathcal{C} = \frac{1}{2\pi} \oint_{S^2} \boldsymbol{\Omega}_n(\boldsymbol{k}) \cdot d\boldsymbol{S}} \tag{18}$$

This counts the net Berry flux through any closed surface enclosing the Weyl point.

Numerical Integration

Method:

1. Define spherical coordinates (θ, ϕ) centered at $\boldsymbol{k}_W$
2. Sample $\boldsymbol{\Omega}_n \cdot \hat{r}$ on a 20×40 angular grid
3. Integrate using trapezoidal rule on the sphere surface:

$$\mathcal{C}(r) = \frac{1}{2\pi} \sum_{i,j} \Omega_r(\theta_i, \phi_j) \cdot r^2 \sin\theta_i \, \Delta\theta \, \Delta\phi \tag{19}$$

Convergence test: Verified for radii up to $r = 0.10$ Å^{-1} (within distance to nearest other crossing)
Result: $\mathcal{C} = +1$ to numerical precision for all tested radii

6.2 Nielsen-Ninomiya Constraint

Partner Weyl Points

The Nielsen-Ninomiya theorem requires:

$$\sum_{\text{all Weyl points}} \chi_i = 0 \tag{20}$$

Our finding:

- 4 Weyl points along $\langle 111 \rangle$ directions with $\chi = +1$
- 4 partner Weyl points near W points with $\chi = -1$
- Total chirality: $4(+1) + 4(-1) = 0$ ✓

Located by $40 \times 40 \times 40$ grid scan over full BZ.

7 $k \cdot p$ Theory for Weyl Point

7.1 Two-Band Effective Hamiltonian

Near the Weyl point, project onto bands 2 and 3:

$$\boxed{H_{k\cdot p}(\delta \boldsymbol{k}) = \omega_W \mathbb{1}_2 + v_\parallel \delta k_\parallel \sigma_z + v_\perp (\delta k_x \sigma_x + \delta k_y \sigma_y)} \tag{21}$$

Symmetry Constraints

Why this form?
C_3 **symmetry along** [111]**:**

- σ_z term is C_3-invariant (diagonal in the basis)
- $\sigma_{x,y}$ terms form the 2D irrep of C_3
- Bands 2 and 3 belong to *different* irreps (A and E)
- Therefore $\sigma_{x,y}$ terms cannot mix them at $k_\parallel = 0$
- The crossing is *protected* — no mass term $\propto \sigma_z$ can arise from C_3-preserving perturbations

7.2 Chirality from $k \cdot p$

$$\chi = \text{sgn}(v_\parallel v_\perp^2) = \text{sgn}(v_\parallel) \tag{22}$$

For our case: $v_\parallel > 0$ so $\chi = +1$, consistent with Berry integration.

7.3 Fitting Procedure

$k \cdot p$ Parameter Extraction

Method:

1. Extract LSWT band energies in a window of $\pm 8\%$ of $|\Gamma L|$ centered on Weyl point
2. Fit two-band eigenvalues $E_\pm = \omega_W \pm \sqrt{v_\parallel^2 \delta k_\parallel^2 + v_\perp^2 \delta k_\perp^2}$
3. Minimize RMS deviation using Nelder-Mead simplex

Results at $D/J = 0.32$:

- $t_W \approx 0.55$ (fractional position along $\Gamma \to L$)
- $\omega_W \approx 29$ meV
- $v_\parallel \approx v_\perp \approx 10$ meV·Å (nearly isotropic cone)

Agreement: Direct band-structure fits and $k \cdot p$ fits agree within $\sim 10\%$

8 Surface Arc Calculation

8.1 Slab Geometry

Slab Construction

Setup:

- 50 unit cells stacked along a_3 ([110] direction)
- Open boundary conditions at top/bottom surfaces
- Periodic boundary conditions in the two in-plane directions

Hamiltonian:

- Fourier transform only in 2 periodic directions
- Retain real-space hopping along finite direction
- Results in 200×200 matrix at each $\boldsymbol{k}_\parallel$

Surface spectral weight:

$$w_{\text{surf}}(n, \boldsymbol{k}_\parallel) = \sum_{l \in \text{surface}} |\psi_n(l, \boldsymbol{k}_\parallel)|^2 \tag{23}$$

where sum is over the outermost 2 unit cells on each surface.

Surface Arc Detection

Surface arcs are visible as branches with elevated surface weight that:

- Connect projections of Weyl points onto surface BZ
- Cross the bulk Weyl energy $\omega_W \approx 29$ meV
- Are topologically protected (cannot be removed without closing bulk gap)

9 Magnon Hall Conductivity

9.1 Kubo Formula for Thermal Hall

$$\boxed{\kappa^{xy} = -\frac{k_B^2 T}{\hbar} \frac{1}{V} \sum_{n,\boldsymbol{k}} c_2(n_B(\varepsilon_{n\boldsymbol{k}}))\, \Omega_{n\boldsymbol{k}}^{xy}} \tag{24}$$

Component Definitions

Bose-Einstein distribution:

$$n_B(\varepsilon) = \frac{1}{e^{\varepsilon/k_B T} - 1} \tag{25}$$

Magnon Hall weight function:

$$c_2(x) = (1+x)\left[\ln \frac{1+x}{x}\right]^2 - (\ln x)^2 - 2\,\mathrm{Li}_2(-x) \tag{26}$$

where Li_2 is the polylogarithm.
This function weights the Berry curvature contribution by the thermal occupation.

9.2 Comparison with Experiment

Limitations of LSWT Magnon Hall Calculation

Observed discrepancy: Theory exceeds experiment by factor ~ 200
Reasons for overestimate:

1. Magnon-phonon scattering (not in LSWT) reduces $|\kappa^{xy}|$
2. Magnon-magnon interactions absent
3. Finite magnon lifetime from extrinsic scattering
4. Comparison is with $Lu_2V_2O_7$, not $Y_2V_2O_7$

Qualitative agreement:

- Both theory and experiment show rise from zero at low T
- Confirms Berry curvature from Weyl magnon as dominant source

10 RIXS Formalism

10.1 Kramers-Heisenberg Formula

$$\boxed{\frac{d^2\sigma}{d\Omega\, d\omega_f} = \sum_f \left| \sum_\nu \frac{\langle f|\mathcal{O}^\dagger_{\epsilon_f}|\nu\rangle\langle\nu|\mathcal{O}_{\epsilon_i}|g\rangle}{E_g + \hbar\omega_i - E_\nu + i\Gamma_c/2} \right|^2 \delta(E_f - E_g - \hbar\omega)} \tag{27}$$

Physical Interpretation

States:

- $|g\rangle$: Ground state (energy E_g)
- $|\nu\rangle$: Intermediate state with core hole (energy E_ν)
- $|f\rangle$: Final state (energy E_f)

Process at V L_3 edge:

1. Incident photon ($\hbar\omega_i \approx 518$ eV) excites $2p_{3/2} \to 3d$
2. Intermediate state: $2p^5 3d^2$ configuration
3. Core hole decays: $3d \to 2p$ photon emission
4. Final state: excited $3d$ configuration

Resonance: Cross-section peaks when $E_g + \hbar\omega_i \approx E_\nu$

10.2 Polarization Selection Rules

Channel Separation

$\sigma \to \pi$ **(spin-flip):**

- Effective operator carries $\Delta m = \pm 1$
- Selects $\Delta S_z = \pm 1$ final states
- Contains: Exchange peak (8 meV), SOC peak (22 meV)

$\sigma \to \sigma$ **(spin-conserving):**

- Symmetric operator, $\Delta S_z = 0$
- Contains: Phonon sideband (100 meV)

This polarimetric separation is the **central experimental prediction**.

11 V^{4+}–V^{4+} Dimer Exact Diagonalization

11.1 Hilbert Space Construction

State Counting

Single-site initial states:

$$\binom{10}{1} = 10 \text{ states (one electron in 5 } d \text{ orbitals} \times 2 \text{ spins)} \tag{28}$$

Two-site initial states:

$$10 \times 10 = 100 \text{ states before symmetry} \tag{29}$$

After including exchange coupling and SOC: 190 states
Intermediate states (one core hole):

- Core hole can be on either site
- Configuration: $2p^5 3d^2$ at one site, $3d^1$ at other
- Total: 4560 states

11.2 Hamiltonian Parameters

Parameter	Value	Notes
$10Dq$	1.90 eV	Octahedral splitting
δ_{trig}	30 meV	Trigonal distortion
ζ_{3d}	30 meV	3d spin-orbit coupling
ζ_{2p}	4.65 eV	2p spin-orbit coupling
Γ_c	165 meV (HWHM)	Core-hole lifetime
F^2_{dd}	8.102 eV	Slater integral (80% atomic)
F^4_{dd}	5.083 eV	Slater integral (80% atomic)
J	8.22 meV	Inter-site exchange
U_{dd}	3.0 eV	Hubbard U

Table 2: EDRIXS calculation parameters

Slater Integral Scaling

Atomic Slater integrals from Cowan's code are scaled to 80% to account for configuration interaction screening. This is standard practice but introduces $\sim$ 10–20% uncertainty in peak positions.

11.3 Phonon Coupling

Displaced Oscillator Model

$$\mathcal{H}_{\text{ph}} = \hbar\omega_{\text{ph}} b^\dagger b + g(b + b^\dagger)\hat{n}_d \tag{30}$$

Parameters:

- $\hbar\omega_{\text{ph}} = 100$ meV (V–O breathing mode)
- $g = 35$ meV (linear coupling)
- Huang-Rhys factor: $S_{HR} = g^2/(\hbar\omega_{\text{ph}})^2 \approx 0.12$

Truncation: Phonon Hilbert space limited to 4 quanta (converged at this coupling)

12 Summary of Key Predictions

Observable	Predicted Value	Current Resolution	Detectable?
Weyl energy ω_W	29 meV	30 meV FWHM	Marginal
Exchange peak	8 meV	30 meV FWHM	No (need $<$ 10 meV)
SOC peak	22 meV	30 meV FWHM	Yes
Phonon sideband	100 meV	30 meV FWHM	Yes

Table 3: Experimental detectability at current RIXS resolution

Main Conclusions

1. **Weyl magnon exists** at $\omega_W \approx 29$ meV with $\mathcal{C} = +1$
2. **Protected by C_3 symmetry** — robust to parameter uncertainties
3. **Polarimetric RIXS** can separate magnetic (spin-flip) from lattice (spin-conserving) excitations
4. **SOC and phonon peaks** testable now; exchange peak requires next-gen resolution

13 Potential Questions and Responses

13.1 On LSWT Validity

Q: How can LSWT be trusted for $S = 1/2$?
A: While quantum corrections are nominally $O(1)$, the *topological* aspects are protected:

- Weyl point existence: protected by C_3 symmetry (cannot be gapped)
- Chern number: topologically quantized (cannot change without gap closing)

- Interactions shift ω_W and t_W but preserve the crossing
- Prior $1/S$ studies show $\sim 20\%$ renormalization, not 100%

13.2 On Parameter Transfer

Q: Why use $Lu_2V_2O_7$ parameters for $Y_2V_2O_7$?
A: Best available data. Key mitigations:

- Similar T_C suggests similar J
- DM ratio affects position but not existence of Weyl point
- Topology persists over $D/J \in [0.1, 0.6]$ — robust to $\pm 20\%$ uncertainty

13.3 On Magnon Hall Discrepancy

Q: Why is theory 200× larger than experiment?
A: LSWT gives the *intrinsic* contribution only. Missing physics:

- Magnon-phonon scattering
- Magnon-magnon interactions
- Finite magnon lifetime
- Comparison is with different compound (Lu vs Y)

The qualitative trend (rise from zero, Berry curvature origin) is correct.

13.4 On RIXS Momentum Resolution

Q: Can RIXS actually probe the Weyl point at finite k?
A: Current soft X-ray RIXS achieves ~ 0.03 Å^{-1} momentum resolution, sufficient to map dispersion across significant BZ fraction. The dimer calculation gives the zone-center response; momentum-resolved extension is left to future work.